

Online 3

Operator Overloading

Sub-section: B1/B2

Time: 45 minutes for coding + 5 minutes for submission

Marks: 10

Implement a C++ class named `ShiftArray` that manages a dynamically allocated array of integer values and overloads operators for shifting elements and printing.

The class must:

- Store elements in a dynamically allocated integer array
- Overload binary left (`<<`) and right (`>>`) shift operators to shift array elements:
 - Left shift (`array << n`): shifts elements to the left by `n` positions (`n >= 0`); elements shifted out wrap around to the end (circular shift)
 - Right shift (`array >> n`): shifts elements to the right by `n` positions; elements shifted out wrap around to the beginning (circular shift)
- Overload compound shift operators (`<<=` and `>>=`):
 - `array <<= n` performs a left circular shift by `n` positions in place
 - `array >>= n` performs a right circular shift by `n` positions in place
- Provide a `print()` member function to display the array contents in `[ x1, x2, x3, ... ]` format
- Provide a constructor that initializes the array from a raw `float` array
- Perform deep copying and proper memory cleanup

No STL containers are allowed.

Example Code

```
#include <iostream>
using namespace std;
int main() {
    float init[] = {1.0f, 2.0f, 3.0f, 4.0f, 5.0f};
    ShiftArray arr(init, 5);
    // Left shift by 2 positions
    ShiftArray leftShifted = arr << 2;
    cout << "Left shifted by 2: " ;
    leftShifted.Print();
    cout << endl;
    // Right shift by 3 positions
    ShiftArray rightShifted = arr >> 3;
    cout << "Right shifted by 3: ";
    rightShifted.Print();
    cout << endl;
    // In-place shifting using compound operators
    arr <<= 1;
    cout << "After <<= 1: ";
    arr.Print();
    cout << endl;
    arr >>= 2;
    cout << "After >>= 2: ";
    arr.Print();
    cout << endl;
    return 0;
}
```

Expected Output

```
Left shifted by 2: [3, 4, 5, 1, 2]
Right shifted by 3: [3, 4, 5, 1, 2]
After <<= 1: [2, 3, 4, 5, 1]
After >>= 2: [5, 1, 2, 3, 4]
```